Bent-crystal monochromators with xrt 2.0 and OASYS2

Porting, bug fixes, reruns and next steps

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`github.com/oasys-esrf-kit/bent_crystals_llnl_esrf`
`github.com/oasys-esrf-kit/OASYS2-XRT`

LNLS XRT: four sagittally focusing bent-crystal monochromators

Beamline	Source	Monochromator	Energies
APS_BLM	7 GeV BM, 0.6 T	Si(311) Bragg (Johann) + bent Laue	30–60 keV
SSU_BLM	3 GeV BM, 6.6 T	Si(311) Bragg (Johann) + bent Laue	35–200 keV
SSU_SBDL	3 GeV BM, 6.6 T	two bent Si(111) Laue (PP)	35–200 keV
BNL_SBDL	3 GeV wiggler, $K = 16.5$	two bent Si(111) Laue (ML) + Pt mirror	30–70 keV

- One Python script per beamline (`XRT_*.py`) scans the energy. At each energy:
  - Bragg angle, crystal positions, analytic radii R_s (thin lens) and R_m (anticlastic)
  - optimization of `Fcurv`, a scale factor on the sagittal bending, to focus on the sample
  - full trace (500k rays): flux, sizes, divergences, bandwidth, focus positions
- Crystal reflectivity: use an extension `BentCrys2D.py` to compute the reflectivity of bent crystals using the Penning–Polder (PP, Laue only) or multilamellar (ML) model

What was done

Software

- OASYS1-XRT ported to **OASYS2-XRT** (`https://github.com/oasys-esrf-kit/OASYS2-XRT`)
- New widgets: Bent Crystal 2D (bent Laue/Bragg Si), Bending Magnet, Wiggler, Johann Cylinder crystal
- `BentCrys2D.py` adapted to xrt 2.0 (no patched `oes_base.py` needed)
- Bragg angle from xrt instead of optlnls (≤ 1.4 μrad difference)

Simulations

- All scripts run with xrt 2.0, full energy scans
- One OASYS2 workspace per beamline at a single energy (`oasys_*/`, with `INFO.md`)
- Internal reflectivity checked against XOPPY `diff_pat`
- Two bugs found and fixed, all scans rerun
- Repository `https://github.com/oasys-esrf-kit/bent_crystals_llnl_esrf` with code, runs, READMEs and slides

How the bent-crystal model works

- **Direction:** each ray is diffracted like a grating (surface component of **k** changes by the surface component of **H**), using the local plane and surface normals of the bent crystal. Focusing comes from R_m and $R_s \cdot$`Fcurv`.
- **No penetration:** rays diffract and leave at the entrance surface. The thickness enters only the reflectivity (no Borrmann fan, $\sim$0.1 mm for 1 mm Si at 35 keV).
- **Reflectivity, per ray:**
  - local deviation from the Bragg condition, asymmetry from the actual directions
  - strain gradient from the bending moments and the anisotropic compliance tensor
  - **PP** (Laue): integration of the deviation between entrance and exit faces
  - **ML**: lamellae summed incoherently, weighted by transmission and absorption (from `crystal3.f90`, as in XOPPY)
- **Depth reference:** the Bragg-condition point is at the crystal *mid-plane*; `PP_center_fraction` is stored but ignored. XOPPY uses the entrance surface. This matters later.

Bug 1: local beam passed downstream of the 10 m mask

- `propagate()` and `expose()` return the beam in the *local* frame ($y = 0$). The scripts passed it to the next element as if it were global.
- Effect: all optics sat **10 m farther from the source**. The `Fcurv` optimization compensated, which is why `Fcurv` was 1.05–1.20 everywhere.
- Fix: pass the global beam (`beamSource`); the mask has already flagged the lost rays.

SSU_BLM at 80 keV, beam at the 56 m screen:

Beam into Bragg	`Fcurv`	x FWHM	horizontal focus	z FWHM
local (script)	1.1175	28 μm	56.00 m	4.0 mm
local (script)	1.0	374 μm	52.25 m	5.1 mm
global (correct)	**1.0**	**40 μm**	**56.00 m**	**2.9 mm**

The OASYS2-XRT Runner always used the global beam: the workspaces were correct.

Bug 2: multilamellar (ML) reflectivity

Three bugs in `bentSi_Crys.get_amplitude`, fixed by comparison with `crystal3.f90`:

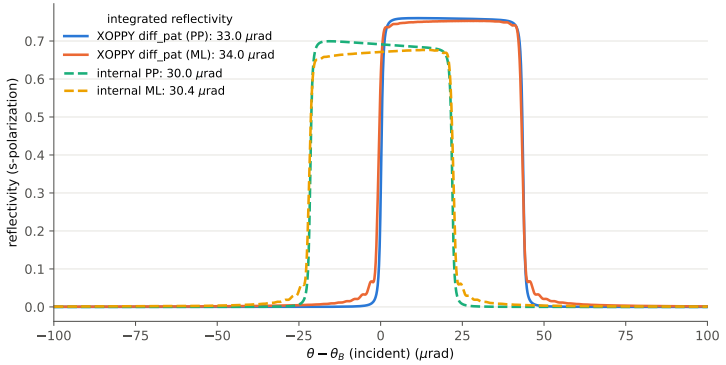
1. Laue: transmission of the lamellae in front never updated (`tsmall_old`) $\Rightarrow R > 1$, capped at 1
2. Lamella step refined $\times 4$; ML is incoherent, so it does not converge. Now $\Delta y = \pi/2$ (Laue), 2 (Bragg)
3. xrt returns the complex conjugates of $\chi_0, \chi_H, \chi_{\bar{H}} \Rightarrow$ absorption acted as gain; thick Bragg crystals reflected $R \approx 1$ over 330 μrad

Integrated s-reflectivity at 80 keV (μrad), Si(311):

Case	XOPPY ML	before	fixed	fixed + XOPPY μ
Laue, SSU_BLM radii	34.02	48.77	30.42	34.04
Laue, low R	62.85	60.12	56.35	63.17
Bragg, SSU_BLM	67.45	315	17.1	71.4

Remaining difference = **absorption coefficient**: xrt total $\mu = 0.521$ cm^{-1} vs XOPPY photoelectric 0.118 cm^{-1} (Si, 80 keV). Matters $\times 4$ for thick Bragg crystals.

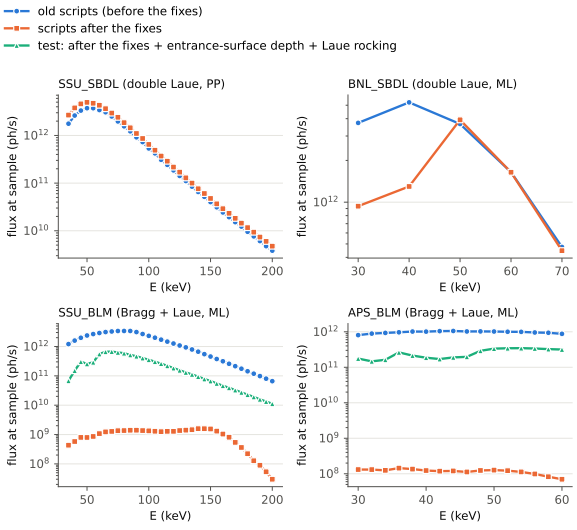
Laue rocking curve: internal code vs XOPPY (after the ML fixes)



SSU_BLM Laue crystal: Si(311), 80 keV, asymmetry 20.98°, $t = 1$ mm, $R_s = 1217.6$ mm, $R_m = -76039$ mm (`XOPPY/comparison_SSU_BLM.py`).

- Internal PP and ML now agree (PP/ML = 0.988; XOPPY 0.969), about 11% below XOPPY (absorption: xrt uses the total μ, possibly including Compton)
- The internal profiles are centred on the crystal mid-plane, XOPPY's start at the entrance surface: shift of half the width ($\sim$22 μrad), same integral

Reruns: flux at the sample



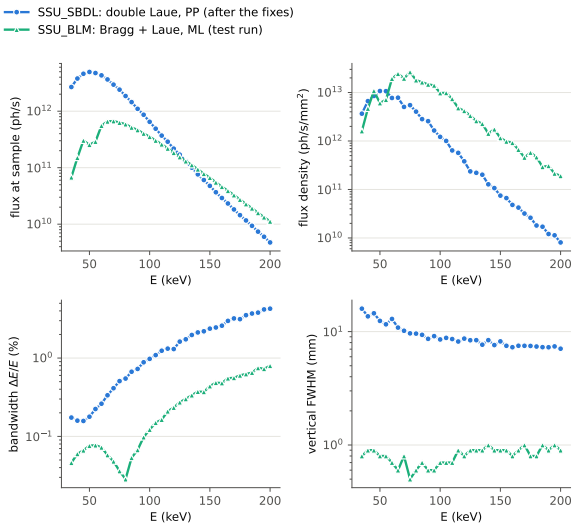
The three curves

- **blue**: old runs, with both bugs
- **orange**: the scripts with the two fixes, otherwise unchanged
- **green** (Bragg+Laue only): orange + two changes *not yet in the scripts*: (1) depth reference at the entrance surface (as XOPPY); (2) at each energy, the Laue pitch is scanned ($-60 \ldots +160$ μrad, 5 μrad steps) and set to the offset of maximum intensity after the Laue crystal; the script then runs normally at that pitch

Valid / invalid is not about the flux level

- **valid**: the simulated monochromator works (crystals aligned, focus found); the numbers can be quoted. SSU_SBDL $\times$1.2–1.5, BNL_SBDL $\times$0.25–1.07
- **invalid** (orange Bragg+Laue curves): in the model the Laue crystal, set at the nominal angle, misses the band reflected by the Bragg crystal. Only 10^{-4}–10^{-3} passes; sizes, bandwidth and `Fcurv` come from a few rays. An alignment artefact, not a property of the beamline

SSU: double Laue (SBDL) vs Bragg + Laue (BLM)



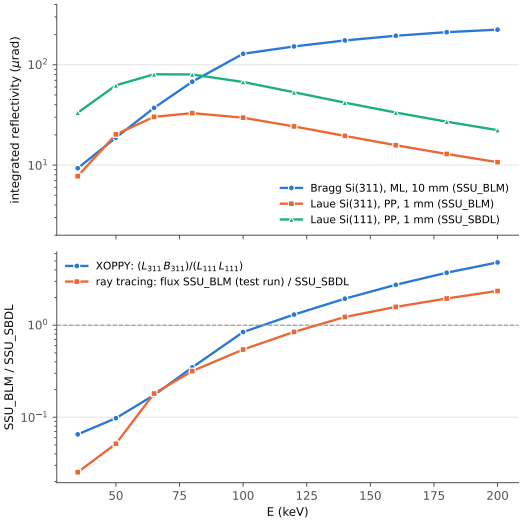
Same source (3 GeV BM, 6.6 T) and same sample screen (56 m).

- **Flux:** SBDL is higher up to 130 keV ($\times$14–40 at 35–55 keV, $\times$3 at 80 keV); BLM is higher above 130 keV ($\times$2.4 at 200 keV)
- **Bandwidth:** BLM is 2–20$\times$ narrower ($\Delta E/E$ 0.03–0.8% vs 0.16–4.3%), which accounts for much of the flux difference
- **Vertical size:** BLM 0.5–1 mm (the Johann Bragg crystal focuses vertically), SBDL 7–16 mm (no vertical focusing)
- **Flux density:** so BLM wins from 60 keV on, by $\times$2.5–25
- **Horizontal FWHM:** similar, 34–84 μm (SBDL) and 26–60 μm (BLM)

Caveats

- SSU_BLM is the *test run* (entrance-surface depth + Laue rocking), not the script. It depends on the open decisions, in particular the absorption coefficient (up to $\times$4 on the Bragg reflectivity)
- SBDL uses PP for its Laue crystals, BLM uses ML (they agree within about 1%)

Why SSU_BLM wins at high energy (XOPPY)



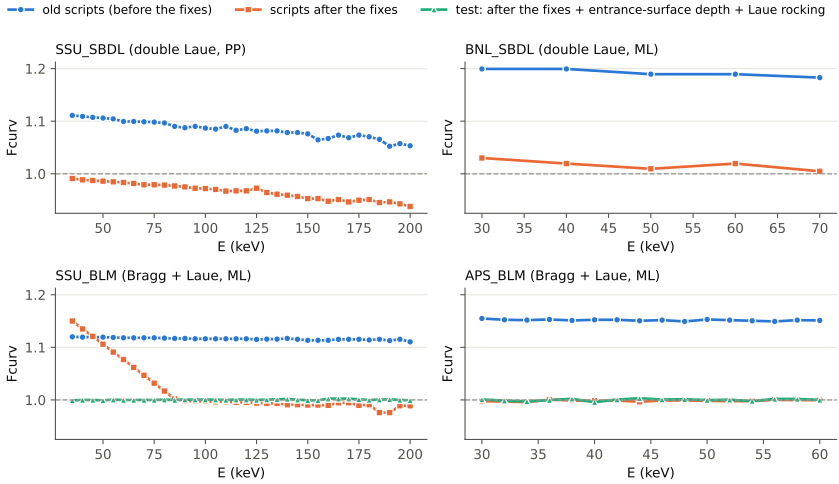
XOPPY script `XOPPY/integrated_SSU.py`.

- **Laue, 1 mm:** peak reflectivity falls from ∼0.9 (65 keV) to 0.08–0.09 (200 keV); the integrated reflectivity falls above 65–80 keV
- **Bragg, 10 mm bent:** peak stays ≈1 and the integrated reflectivity grows from 9 to 224 μrad
- So SSU_SBDL pays the Laue loss twice, SSU_BLM once
- **Figure of merit** $(LB)/(LL)$: crosses 1 at ∼110 keV; the ray tracing crosses at ∼130 keV, within a factor ∼2 everywhere

Caveats

- The product ignores how the two curves overlap for each ray
- XOPPY uses the photoelectric μ; the ray tracing uses xrt's total μ, which lowers the Bragg reflectivity (×0.25 at 80 keV)
- The large SSU_SBDL bandwidth is an energy–angle spread over the beam (0.66 mrad $\times \cot\theta_B$), not throughput: at 200 keV only 0.05% of its 4.3% band gets through on average (SSU_BLM: 0.6% of 0.8%)

Reruns: `Fcurv` is now ≈ 1



With the correct geometry the analytic radii focus on the sample (SSU_SBDL 0.94–0.99, BNL_SBDL 1.005–1.03, test runs 0.996–1.003). SSU_BLM after the fixes: the search fails at low energy (no flux).

Bragg + Laue: the reflection windows no longer overlap

Diagnosis

- Before the ML fix the Bragg crystal reflected over $\sim$330 μrad: any Laue alignment worked
- Now its window is narrow, and the mid-plane depth reference shifts it by $\sim$160 μrad in the 10 mm crystal
- The scripts never rock the Laue crystal
- Double Laue is weakly sensitive: -20% (SSU_SBDL), $+4\%$ (BNL_SBDL)

Test runs (`test_fc0_rock/`)

- `PP_center_fraction`$= 0$ (entrance surface): profiles match XOPPY within 0.1 μrad
- Before each energy: Laue pitch scanned $-60 \ldots + 160$ μrad (5 μrad steps, 100k rays) and set to the maximum of the intensity after the Laue crystal; Bragg crystal untouched
- SSU_BLM: best offset $0 \rightarrow +120$ μrad with energy, gain up to $\times 30$; APS_BLM: $+5$ to $+30$ μrad
- `Fcurv` ≈ 1, focus on the screen, x FWHM 26–60 μm (SSU_BLM), vertical 0.5–1 mm
- Flux 0.06–0.21$\times$ (SSU_BLM), 0.16–0.36$\times$ (APS_BLM) the old values, mostly due to μ

Summary of results after the fixes

	Peak flux (ph/s)	x size (μm)	z size	`Fcurv`	Status
SSU_SBDL	4.95×10^{12} at 50 keV	34–84	7–16 mm (no focusing)	0.94–0.99	**valid**
BNL_SBDL	3.92×10^{12} at 50 keV	404–463	0.11–0.13 mm	1.005–1.03	**valid**
SSU_BLM	6.7×10^{11} at 70 keV	26–60	0.5–1.0 mm	0.999–1.002	test run
APS_BLM	3.5×10^{11} at 54 keV	167–234	0.45–1.15 mm	0.996–1.003	test run

- The old results (`run_XRT_*/`) are superseded: wrong geometry for all four beamlines, wrong ML flux
- OASYS workspaces rerun with the fixed code: they agree with the scripts (the Bragg+Laue ones fail the same way)

Next steps

Decisions

1. **Depth reference:** implement `PP_center_fraction` in `BentCrys2D.py` and OASYS2-XRT; use 0 (entrance surface, as XOPPY) in the scripts, the widget default and the `.ows` files?
2. **Second-crystal alignment:** add a per-energy Laue rocking scan to `XRT_SSU_BLM.py`, `XRT_SSU_BLM_PP.py`, `XRT_APS_BLM.py`?
3. **Absorption coefficient:** xrt total μ (now) or photoelectric (XOPPY)? $\times 4$ on the Bragg integrated reflectivity.

Open questions and work

4. BNL_SBDL: understand the drop to $0.25\times$ at 30–40 keV
5. Rerun the Bragg+Laue beamlines (and the double-Laue ones if the depth reference changes)
6. Longer term: compare with xrt 1.x results; consider penetration (Borrmann fan) for the 25–60 μm foci

New project: Laue and bent crystals in SHADOW4

Plan

Motivation and starting point

Why: SHADOW4 is the native ray tracer of OASYS2, with Jones-calculus polarization, but it cannot trace any of the four beamlines above: it has no Laue crystals and no bent-crystal reflectivities.

Component	Status
`S4Crystal` (SHADOW4)	Perfect crystals, Bragg only (`BraggDiffraction` hard-coded; Laue "to be developed"); plane, conic, toroid, mesh surfaces
`S4MosaicCrystal`	Separate family of crystal classes: template for a new one
crystalpy	Perfect-crystal amplitudes (Zachariasen, Guigay) for `BraggDiffraction`, `LaueDiffraction`, `LaueTransmission`; susceptibilities from xraylib, DABAX or the SHADOW preprocessors
`bentSi_Crys.get_amplitude`	PP and ML bent-crystal reflectivity per ray, now checked against XOPPY, but tied to xrt objects
XOPPY CRYSTAL (`diff_pat`)	Perfect, PP, ML, Bragg and Laue
XOPPY CRYSTALPY (`crystalpy`)	Perfect Zachariasen, Guigay, Bragg and Laue

Lessons from the xrt work to carry over: check the sign convention of the susceptibilities, use the entrance-surface depth reference, validate every model against XOPPY before tracing a beamline.

Steps i and ii: bent reflectivities, and bent Bragg in `S4Crystal`

i) PP and ML reflectivities on a crystalpy setup

- Extract the physics of `bentSi_Crys.get_amplitude` into a standalone module, independent of xrt: inputs are the per-ray energy, $\hat{k}_{\mathrm{in}}$, $\hat{k}_{\mathrm{out}}$, local Bragg-plane and surface normals, thickness, R_m, R_s, compliance tensor, model (PP, ML)
- Take d, χ_0, χ_H, $\chi_{\bar{H}}$ from a crystalpy `DiffractionSetup` (xraylib or DABAX); **verify their sign convention** (the xrt ones had to be conjugated)
- Implement `PP_center_fraction` (depth reference), default entrance surface
- Validate: rocking curves and integrated reflectivities vs XOPPY `diff_pat` (SSU_BLM and SSU_SBDL crystals, 35–200 keV) and vs the xrt implementation

ii) Bent Bragg (ML only) in `S4Crystal`

- New option for the reflectivity model: perfect (crystalpy, as now) or bent ML
- Directions stay as now (crystalpy, local normal of the surface); only r_σ, r_π change: $\sqrt{R}$, no phase (incoherent model)
- Bending radii given explicitly (they need not equal the surface curvature, cf. `Fcurv`); cylinder/toroid surfaces for the shape
- Test case: SSU_BLM Bragg crystal (Si(311), 10 mm, $R_m = 647.7$ m); compare with XOPPY ML and with the xrt Johann cylinder

Steps iii and iv: `S4LaueCrystal` and penetration

iii) New `S4LaueCrystal` (family of classes for plane, cylinder, toroid, as for mosaic crystals)

- Laue alignment and geometry: beam enters the front face, diffracted beam leaves the back face
- Reflectivity models:
  - perfect crystal: crystalpy `LaueDiffraction` (Zachariasen/Guigay), as SHADOW3 did for Laue crystals
  - bent crystal: PP and ML from step i
- Validate: vs XOPPY (perfect, PP, ML), vs SHADOW3 for perfect Laue, and full beamlines SSU_SBDL and SSU_BLM vs xrt

iv) Penetration in `S4LaueCrystal`

- Today (xrt with the `BentCrys2D` extension) rays diffract and leave at the entrance surface. The real beam leaves from the back face, spread over the Borrmann fan ($2t\tan\theta_B$ symmetric: $\sim$0.1 mm at 35 keV for 1 mm), comparable to the 25–60 μm foci
- Study: where to place the exit point? This influences the "polychromatic focusing"
- Include it as an option and quantify its effect on the focal sizes (SSU_SBDL, SSU_BLM)

Plan: order, deliverables and checks

Step	Deliverable	Check	Depends on
i	bent reflectivity module (PP, ML), crystalpy setup	XOPPY `diff_pat`, xrt	–
ii	`S4Crystal` with bent ML Bragg	XOPPY ML, xrt Johann	i
iii	`S4LaueCrystal`: perfect, PP, ML	XOPPY, SHADOW3, xrt	i
iv	penetration option in `S4LaueCrystal`	focal sizes with/without	iii
–	OASYS2 widgets, example workspaces	the four beamlines	ii, iii

- The open decisions of the xrt study (depth reference, second-crystal alignment, absorption coefficient) apply here too and should be settled first
- End point: the four beamlines traced in both xrt and SHADOW4, with the same crystal physics, as a cross-check of both codes